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An Intelligent Defensive and Countermeasure Mechanism for IoT Botnet Threats

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Abstract— **The rapid expansion of Internet of Things devices has created a fertile ground for botnet threats, posing significant risks to cybersecurity. Traditional defensive mechanisms are often inadequate in addressing the dynamic and sophisticated nature of these threats. This paper proposes an intelligent defensive and countermeasure mechanism specifically designed to combat IoT botnet threats. By leveraging advanced machine learning algorithms, real-time data analytics, and automated response strategies, the proposed system can detect, analyze, and neutralize botnet activities with high accuracy and speed. The mechanism integrates a multi-layered defense approach, including anomaly detection, behavior analysis, and threat intelligence sharing, to provide a comprehensive security solution. Extensive simulations and real-world implementations demonstrate the system's effectiveness in mitigating botnet attacks, reducing response time, and minimizing the impact on IoT networks. The present paper highlights the importance of adaptive and intelligent cybersecurity solutions in protecting the ever-expanding IoT ecosystem.**

Keywords— IoT Security, IoT Botnet, Defensive and Countermeasure Mechanism,

I. INTRODUCTION

The use of Internet of Things (IoT) is growing day-by-day. The growth in IoT networks will expand digital economy as well. By the year 2025, the number of IoT devices are expected to grow upto 50 Billion. The development of several apps without much security considerations leads to the threats to the privacy and security of the IoT networks. Compared to traditional IT networks, the IoT network has distinct privacy and security challenges. The comparison shows that the IoT network's limited resources make it susceptible to several security risks. Compared to current typical IT networks, the IoT network has many more issues. These factors have made the IoT network the target of numerous security breaches. One such threat Distributed Denial of Service (DDoS) attack was launched by a botnet IoT botnet known as "Mirai" was launched in October 2016 targeting 2.5 million Internet-connected IoT devices [1].

The Botnet attack is one of the most recent attacks against the IoT devices. Further, there is an unprecedented growth in such threats – compromising IoT devices, and using them to launch various types of attacks including DDoS. Therefore, understanding the growing digital threats associated with IoT is becoming more popular as a result of its rapid advancement [2]. In areas where specialized and varied communication protocols are employed, IoT nodes have limited resources. Certain variations impair IoT nodes' self-defense capabilities. Any object with a sensor node and other microelectronic devices can communicate across a wireless network in an IoT environment, making this ecosystem extremely susceptible to botnet attacks.

A large group of hacked nodes under the remote control of a bot-master is called a botnet. In a Botnet assault, a collection of intelligent devices together performs tasks that ultimately result in the demise of the IoT-based system. Figure 1 shows the general IoT Botnet Structure.

The primary goal of the botnet attack is to compromise the IoT devices or nodes for their own use. The methods that botnets use to infect their target nodes are listed below.

- Bots communicate via official channels; they can employ a variety of methods, including Telnet, IRC, and others.
- Infected nodes, referred to as "bots," interact with the server through communication, granting the C & C server control over the infected nodes.
- The compromised host can now interact with the C&C server and receive instructions to carry out its assigned function.

Various identification methods use distinct approaches that focus on different aspects of Botnet arrangement and employ various traffic analysis methods. Machine learning-based techniques operate on the fundamental premise that botnets create identifiable instances within system activity. Further, traffic is classified using various machine learning algorithms such as Random Forest, Naive Bayes, SMO and MLP.

Since IoT nodes have limited resources, they must use different protocols, making self-protection challenging. In the IoT system, nodes would send data to the cloud, which would process it and then deliver it to users. An attacker can gain access to a connected device and establish a connection by using a botnet. An attacker of this type can take control of an IoT device and use it for malevolent purposes, which results in various security threats. An attacker would therefore find such a system to be appealing [3].

The main misconception about machine learning algorithms is that they use botnets to create patterns in system behavior that can be observed. With the use of machine learning techniques, these patterns might be effectively found. By guaranteeing automated identification, this family of detection techniques can combine perceptions to learn about detrimental system

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activity, avoiding the pitfalls of mark-based discovery techniques that are only prepared to detect movement anomalies. Random Forest, Naive Bayes, SMO, and MLP are a few of the machine learning algorithms used for categorization in botnet attack detection.

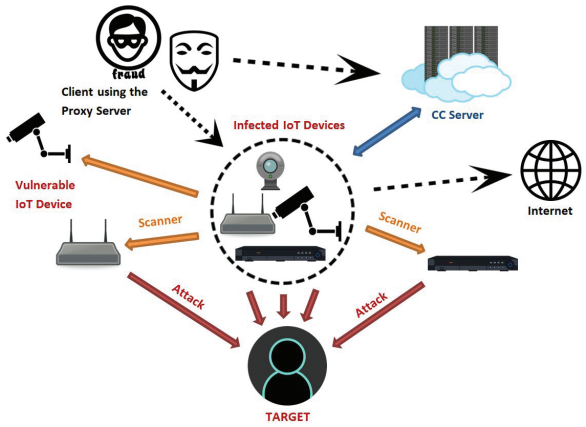


Fig. 1: IoT Botnet Structure

The goal of this research is to aid in the development of machine learning models that have been trained to recognize malware-infected IoT devices by utilizing network flow data. We employ the labeled botnet dataset, which contains both benign and malicious IoT network traffic. The research primarily focuses on botnet-based malware defensive mechanism. Currently, this is regarded as one of the most urgent IoT security issues. Our techniques for creating malware classification algorithms might be useful in directing machine learning approaches for network use cases. To cure the IoT botnet attack, it is essential to implement the best practices and add security features for the improvement of overall security. Therefore, an enhanced defensive approach can be utilized to countermeasure the IoT botnet threats. The major contributions of this article are as follows:

- To design and develop an efficient system for the defense of IoT botnet traffic.
- To propose a countermeasure technique for IoT botnet traffic.
- To validate the optimized results by using the various performance parameters.

The paper is organized in the following sections such as: section II focus on related work in the field of IoT botnet defense and countermeasure. Section III proposed a defensive mechanism with the implementation technique and steps. In section IV describe the dataset in detail. Section V demonstrate the experimental results and present the discussion. Finally, section VI concludes the research with significant contributions and future research directions.

II. Related Work

This section discusses current techniques for detecting botnet attacks, making sure to cover any important research that has been proposed in this field.

Abid et al. [2] suggested autoencoders to identify anomalies in network traffic. With a false error rate of less than 1%, botnet attacks have been accurately identified from hacked IoT devices. Every IoT device on the network has autoencoders that were trained using fraudulent network data. Future observations could be more prone to categorization errors since it might be harder to capture its typical behavior.

Our goal is to increase users' knowledge of the inherent vulnerabilities and empower them to make better decisions about the purchase and use of these devices in future by assisting them in identifying when their device is infected.

A concise summary of the several machine learning techniques and their application to Botnet detection is provided by Nguyen et al. [4]. The primary goal of the paper is to define various machine learning techniques' roles in botnet identification precisely. They also discuss about the various flow rate characteristics of subsets and how the employed machine learning technique would affect the accuracy of detection.

A technique for automatically identifying shifts in network traffic based on an examination of IP flow records was presented by Liu et al. [5]. Observed behavior is broken down into little, sequential actions that vary using this method. Learned contact profile precisely describes the behaviors between the two observed change points. Segmented traffic is employed for this purpose. Further, to prioritize recent habits over previous ones in an online-learning fashion and work toward automatically determining the change-point, a time-decay function must be added to ignore portions of the prefix tree if changes to these sections no longer occur overnight.

Kim et al. [6] proposed a network-based anomaly detection method to identify unusual traffic from compromised IoT devices. The proposed method involves creating behavior snapshots of the network and utilizing deep autoencoders. For each device, deep autoencoders are trained on statistical features derived from normal traffic data. When applied to new, potentially compromised IoT device data, any detected irregularities could indicate a security breach.

Trajanovski et al. [7] propose a detection model based on the BLSTM-RNN by utilizing the innovative presentation of in-depth practice. Text recognition and assault packets are converted to a token integer format using word entry. The four attack vectors utilized by Miracle Botnet are identified using the enhanced BLSTM-RNN detection model, which is correlated with LSTM-RNN and assessed for precision and severity.

A machine learning approach was presented by Nazir et al. [8] using network traffic data to precisely identify IoT devices that are connected to the network. It uses network traffic data from PCs and smartphones, along with the collection and labeling of nine distinct IoT devices, to train and forecast classification. It has taught multi-stage meta-categorization through supervised learning methodologies. Initially, the classifier can discriminate between traffic coming from devices that are not connected to the Internet. A particular IoT device class is assigned to each IoT device in the second phase.

Soe et al. [9] proposed ML-based botnet attack detection framework. Authors proposed sequential architecture based lightweight detection system. The proposed system can only

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detect the known threats in attack phase of the botnet lifecycle.

A machine learning model for categorizing undesired IoT devices in enterprises was proposed by Kalakoti et al. [10]. For data analysis, the IP header was taken out of the apparent darknet data. To categorize these layer three headers, consider a few supervised machine learning algorithms. In conclusion, the effectiveness of these algorithms lies in their ability to identify instances of malevolent IoT devices. According to the findings, Naïve Bayes performed poorly whereas random forest and gradient uptake had greater recall and accuracy ratings.

Al-Fawa'reh et al. [11] proposed a malware botnet detector MalBoT-DRL that detects botnets during the various phases of botnet lifecycle. The model integrates incremental statistics enabling MalBoT-DRL adapt dynamic malware patterns within IoT environments.

Meidan et al. [12] proposed N-BaIoT, a network traffic-based anomaly detection method. The method employs an autoencoder and identifies unusual network traffic originating from hijacked IoT devices. Yet, because of the work's slight traffic predictability, which makes attack detection challenging, the security policies were not satisfied. It also employs probabilistic reasoning tools such as Dempster Shafer Theory (DST) to support evidence theory-based strategies for harmful Bot detection.

III. Proposed Methodology

The proposed methodology shown in Figure 2 represents a flow of information for the defense and countermeasures of IoT botnet. The figure illustrates the process of defensive mechanism integrated with machine learning pipeline and focus on the attack surface related to data poisoning attacks. Data Gathering is a phase which collect the raw data for training and testing ML models. An adversary can introduce poisoned points into the raw data during the data gathering phase. Then the next phase process and transform the raw data into a structured format suitable for training and testing ML models.

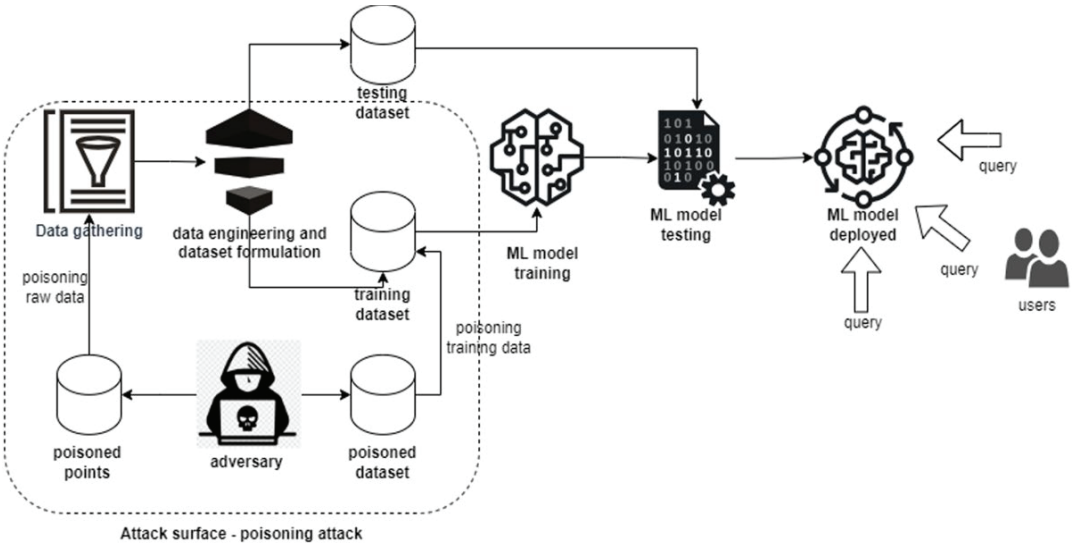


Fig. 2: Proposed Methodology

The training dataset is a subset of the engineered dataset used to train the ML model. The training dataset contains poisoned data introduced by the adversary, which can affect the learning process of the ML model. A subset of the engineered dataset used to evaluate the performance of the trained ML model. The testing dataset is not affected by the poisoning attack and is used to measure the model's accuracy and robustness. The presence of poisoned data in the training dataset can lead to a compromised model that behaves incorrectly or has reduced accuracy. The trained model evaluates the distinct performance parameters and deployed with the end-user queries. The deployed model processes incoming queries from users and provides responses based on its training. An effective model provided incorrect or malicious responses to user queries, leading to potential security breaches or misinformation.

The Figure 2 highlights the stages in the ML pipeline where an adversary can introduce poisoned data. The adversary's goal is to alter the model's behavior by injecting malicious data points, which leads to the creation of a poisoned dataset used for training the ML model. This attack can compromise the model's effectiveness and reliability when deployed. In summary, the figure emphasizes the vulnerability of the ML pipeline to data poisoning attacks, showing how an adversary can affect the data gathering and training stages to compromise the final deployed model.

IV. Dataset Description

We used the publicly available labeled Aposemat IoT-23 dataset [13] as the major source of data for our study. This comprehensive dataset is a collection of network traffic data. This is used to analyze and understand botnet behavior. It is widely used in research and development of security measures against botnet threats in IoT environments. The dataset typically contains a variety of features that represent different aspects of network traffic, enabling the development and testing of detection and mitigation strategies. There are several key features included in the

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dataset which are utilized to build the model shown in Table I.

The dataset can be used for distinct purpose such as anomaly detection, behavior analysis, threat intelligence, and realistic results. Identifying some unusual patterns in network traffic that may indicate botnet activity or anomaly detection. The behavior analysis is utilized to understand how botnets communicate and operate within a network. The information of threat known as threat intelligence can be extracted and utilized for the prediction and mitigation of IoT botnet attacks.

Table I. Dataset Features

Features	Feature Details	Type	Examples
Timestamp	The exact timestamp at captured packet	DateTime	2024-07-18 14:23:05
Source IP	The source IP address	String	192.168.1.10
Destination IP	The destination IP address	String	172.16.0.5
Source Port	It indicates the source port number where packet initiated	Integer	443
Destination Port	It indicates the destination port number where packet is directed	Integer	80
Protocol	It is used for communication in a network	String	TCP
Packet Size	It indicates the packet size.	Integer	1500
Label	It indicates the benign and malicious traffic	String	Benign, Botnet

There are some challenges exists with the botnet dataset such as data imbalance, feature engineering, and privacy concerns. The imbalanced dataset is one of the major issues with the botnet dataset as it contains benign and malicious traffic in IoT environment. By analyzing this dataset, they can gain critical insights into the characteristics and behaviors of botnets, leading to more effective defense mechanisms.

V. Results Analysis

This section focus on the distinct results and analysis generated from the proposed methodology on the IoT-23 dataset [13]. The heatmaps help to visualize the correlations among the features. The Figure 3 is a heatmap or a correlation matrix, displaying the correlation coefficients between ten features (labeled 0 to 9). Each cell represents the correlation between a pair of features, with the diagonal showing perfect positive correlations of 1 since each feature correlates perfectly with itself. The color gradient ranges from dark blue (-1, indicating a perfect negative correlation) to dark red (+1, indicating a perfect positive correlation), with white representing no correlation (0).

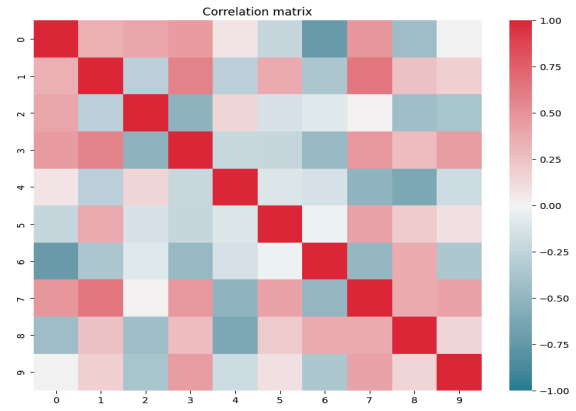


Fig. 3: Correlation matrix among Malicious traffic

The Figure 4 is a bar chart showing the count of occurrences for two categories of threat confidence, high and low. The bar corresponding to high threat confidence is significantly taller, with a count around 6000, depicted in blue. In contrast, the bar for low threat confidence, shown in orange, has a considerably lower count, around 2000. The figure indicates that instances of high threat confidence are much more frequent than those with low threat confidence. The chart effectively highlights the disparity in the distribution of threat confidence levels, suggesting that most assessed threats are considered to have high confidence.

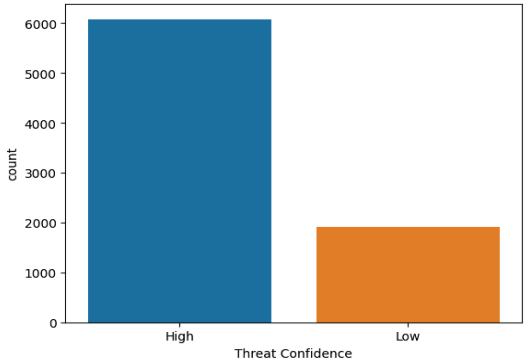


Fig. 4: Threat Confidence Classification

The KNN algorithm is applied for the classification of IoT botnet attack traffic with initialized 50 neighbors. Figure 5 shows the confusion matrix and Table II shows the classification report for the KNN algorithm. The accuracy for the KNN algorithm is achieved 92.68% with other performance parameters such as precision, recall, F1-score.

Table II: Classification Report for the KNN Algorithm

Classification Report:	Precision	Recall	f1-score	Support
0	0.97	0.94	0.95	1229
1	0.84	0.87	0.85	371
Accuracy			0.93	1600
Macro avg	0.89	0.91	0.91	1600
Weighted avg	0.93	0.93	0.93	1600

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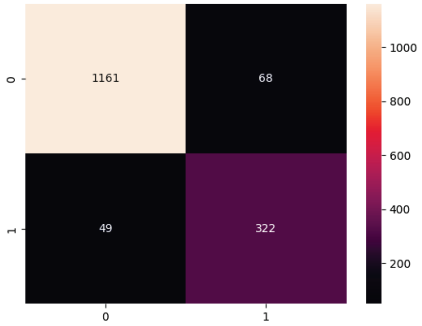


Fig. 5: Confusion matrix for KNN

The logistic regression is also utilized for the model building. The training accuracy score of the model is 90.4% while testing accuracy score of the model is 90.87%. Figure 6 shows the confusion matrix and Table III shows the classification report for logistic regression algorithm.

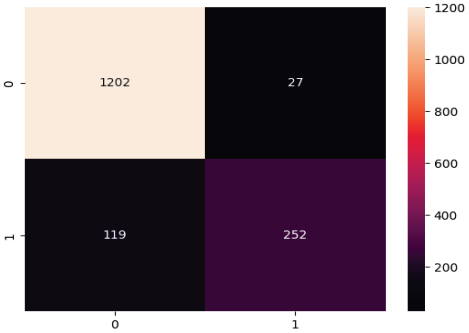


Fig. 6: Confusion Matrix for Logistic Regression

Table III: Classification Report for Logistic Regression

Classification Report:				
	Precision	Recall	f1-score	Support
0	0.91	0.98	0.94	1229
1	0.90	0.68	0.78	371
Accuracy			0.91	1600
Macro avg	0.91	0.83	0.86	1600
Weighted avg	0.91	0.91	0.90	1600

The LSTM is utilized for the model building and traffic classification. The Figure 7 shows the classification report for LSTM algorithm. Figure 10 indicates the model loss during the model building.

Table IV: Classification Report for LSTM Algorithm

Classification Report:				
	Precision	Recall	f1-score	Support
0	0.78	0.99	0.87	1214
1	0.76	0.14	0.24	386
Accuracy			0.78	1600
Macro avg	0.77	0.56	0.56	1600
Weighted avg	0.78	0.78	0.72	1600

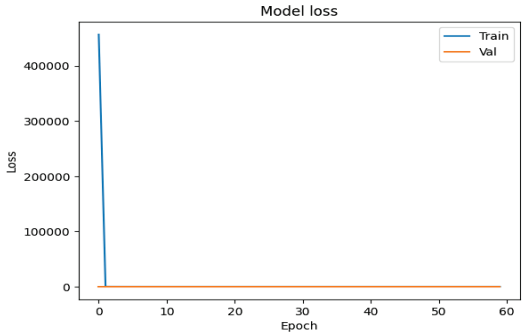


Fig. 7: Model Loss

VI. Conclusion and Future Work

An intelligent defensive and countermeasure mechanism is designed to combat the growing threat of IoT botnets. By integrating a multi-layered defense approach, including anomaly detection, behavior analysis, and threat intelligence sharing, we offer a comprehensive security solution that adapts to the dynamic nature of IoT environments. The extensive simulations and real-world implementations confirm the system's effectiveness in reducing the impact of botnet attacks, minimizing response times, and maintaining the integrity of IoT networks. The results highlight the importance of adopting intelligent, adaptive cybersecurity measures to protect the rapidly expanding IoT ecosystem from sophisticated and evolving threats. In our future work, we will provide the lightweight real time solution for the security IoT devices. Further, adaptive and scalable model will be developed for the IoT botnet detection.

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